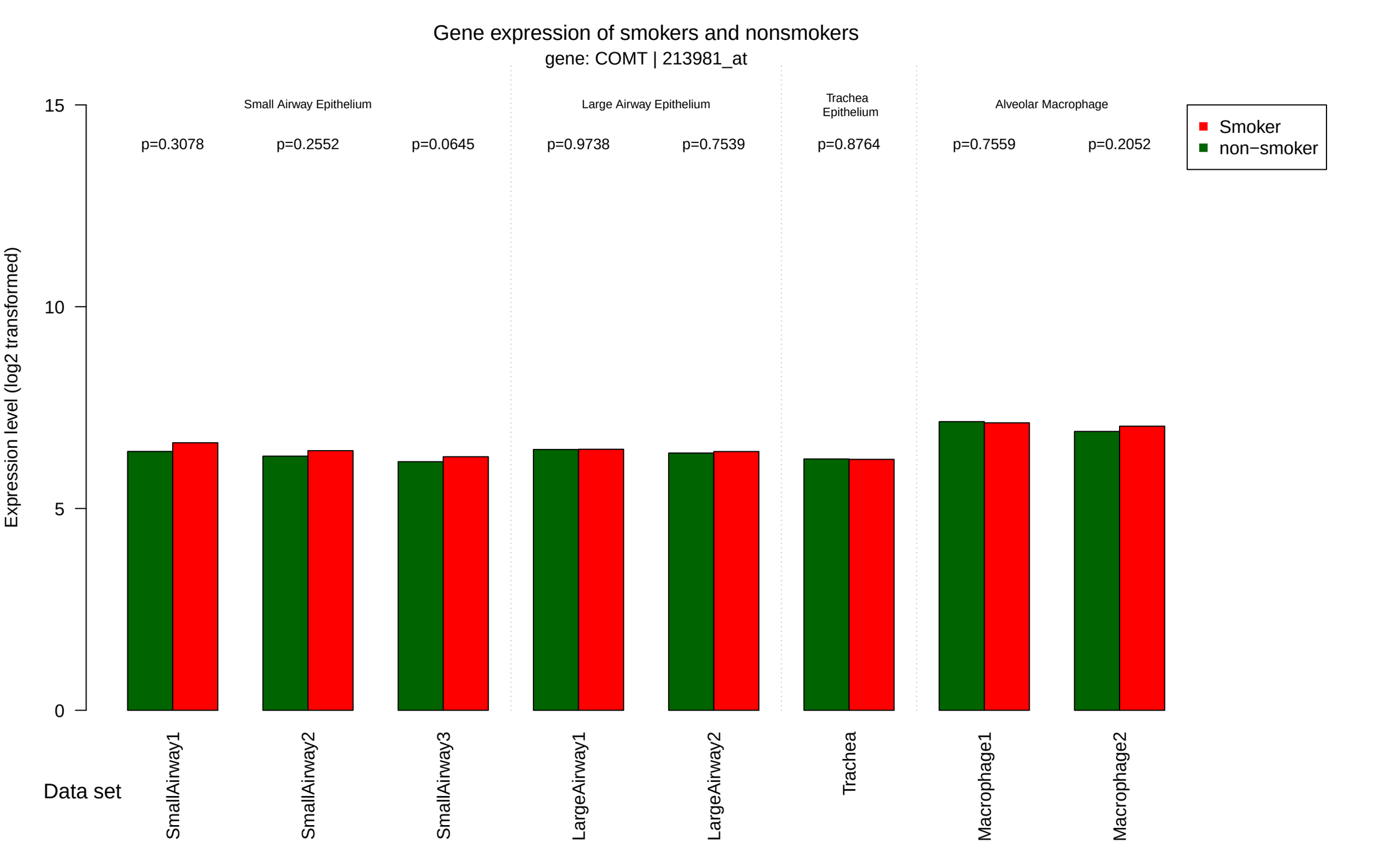
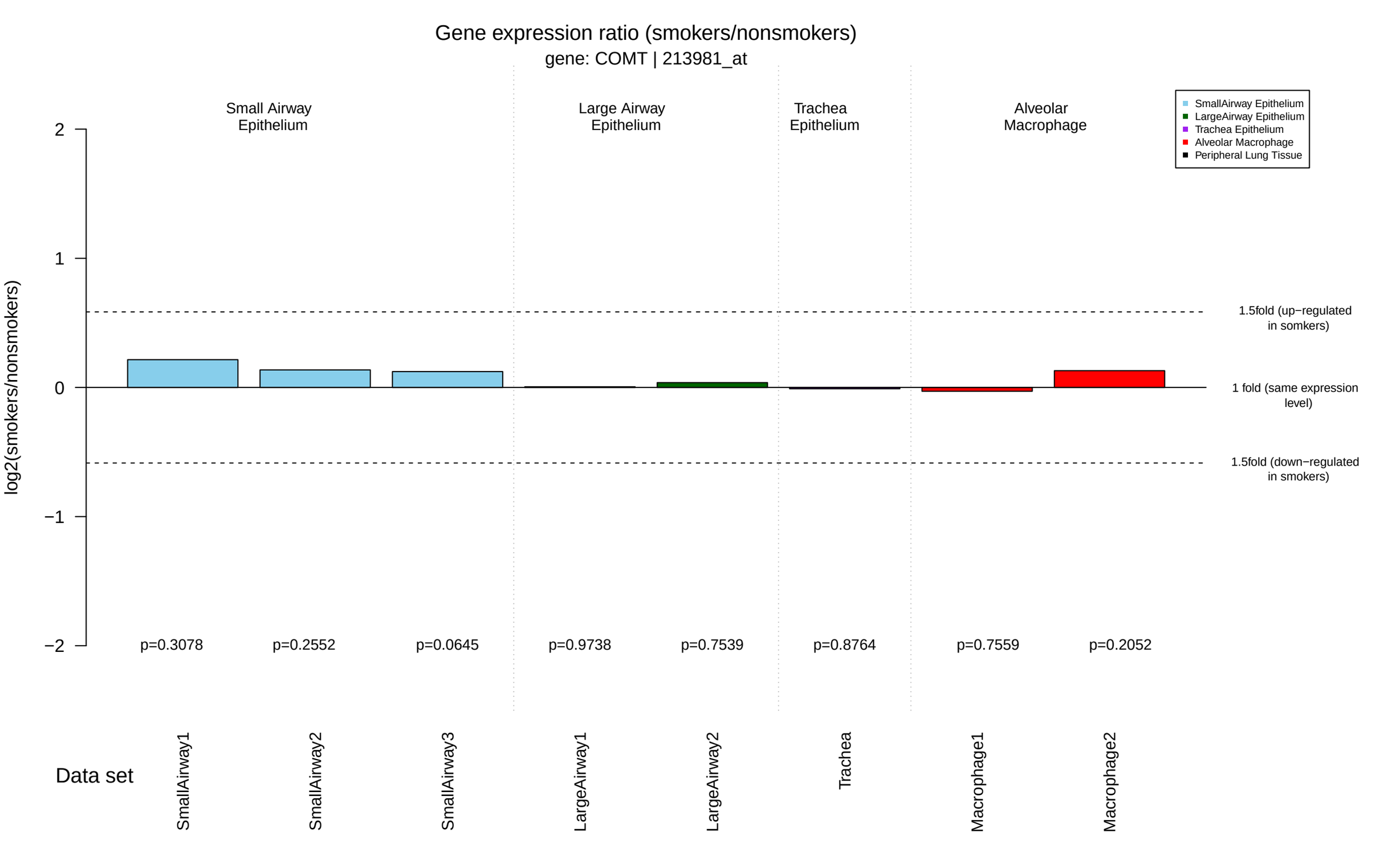
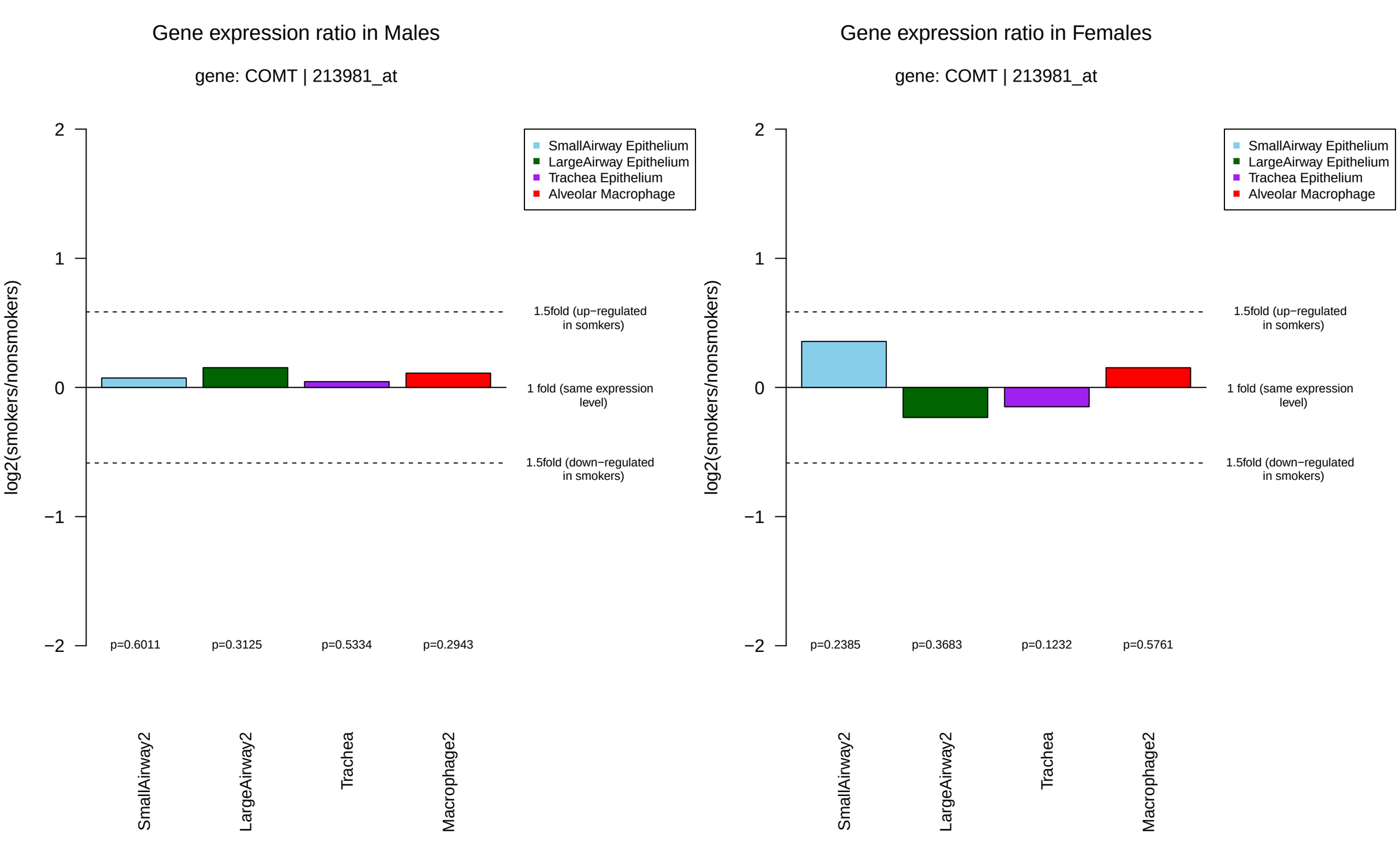
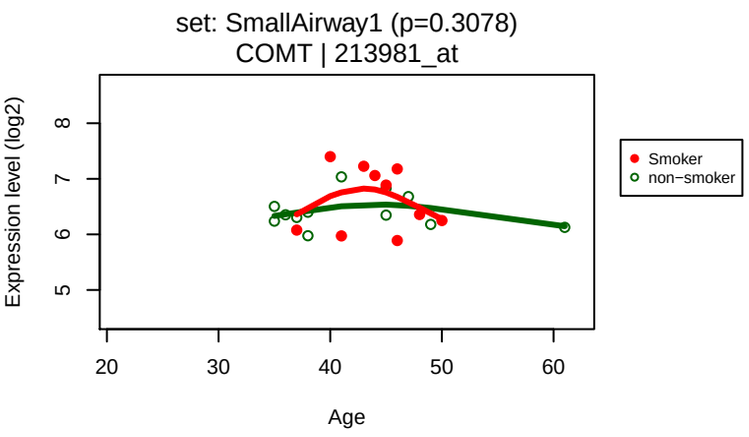
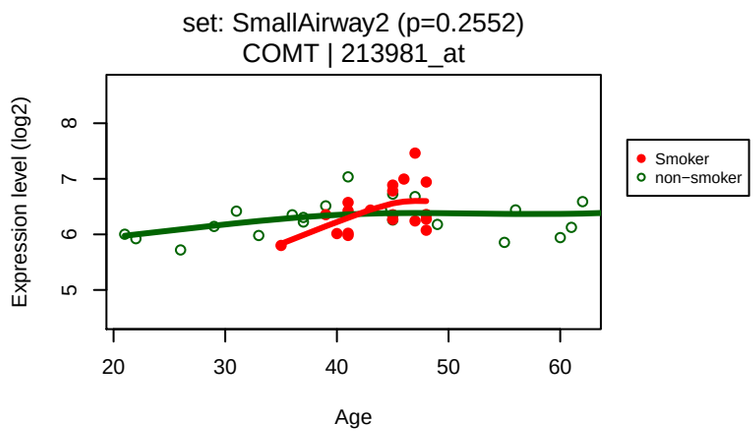
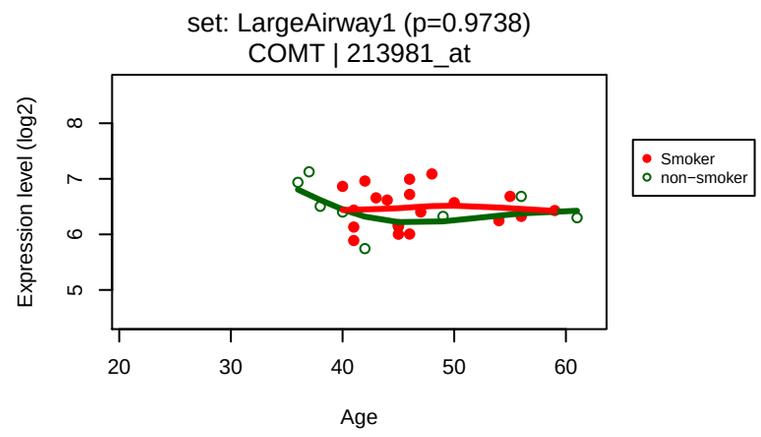

Gene expression of smokers and nonsmokers
gene: COMT | 213981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3078
p=0.2552
p=0.0645
p=0.9738
p=0.7539
p=0.8764
p=0.7559
p=0.2052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

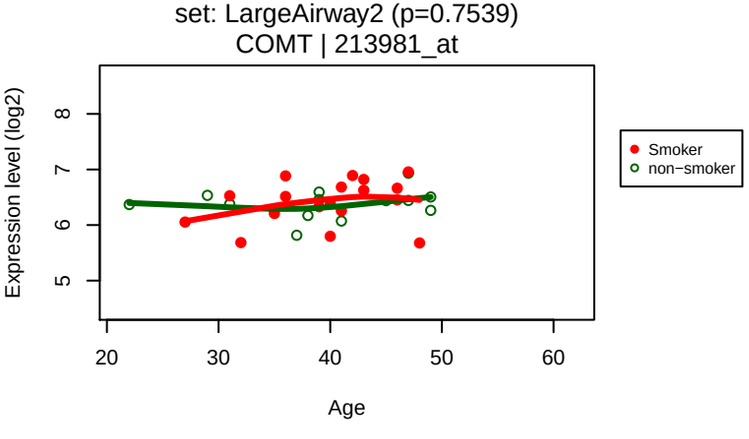

Gene expression ratio (smokers/nonsmokers)
gene: COMT | 213981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3078
p=0.2552
p=0.0645
p=0.9738
p=0.7539
p=0.8764
p=0.7559
p=0.2052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

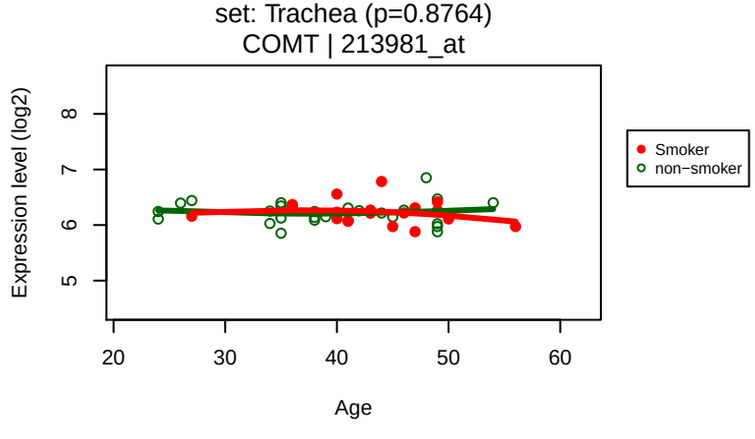

Gene expression ratio in Males
gene: COMT | 213981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6011
p=0.3125
p=0.5334
p=0.2943
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COMT | 213981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2385
p=0.3683
p=0.1232
p=0.5761
SmallAirway2
LargeAirway2
Trachea
Macrophage2

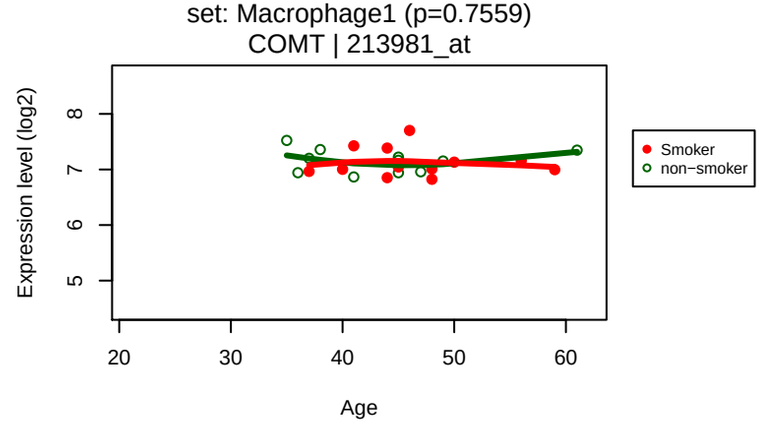

set: SmallAirway1 (p=0.3078)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker

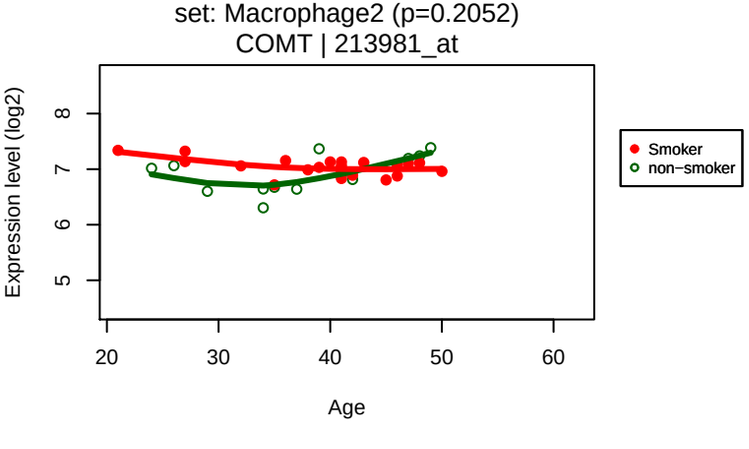

set: SmallAirway2 (p=0.2552)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker

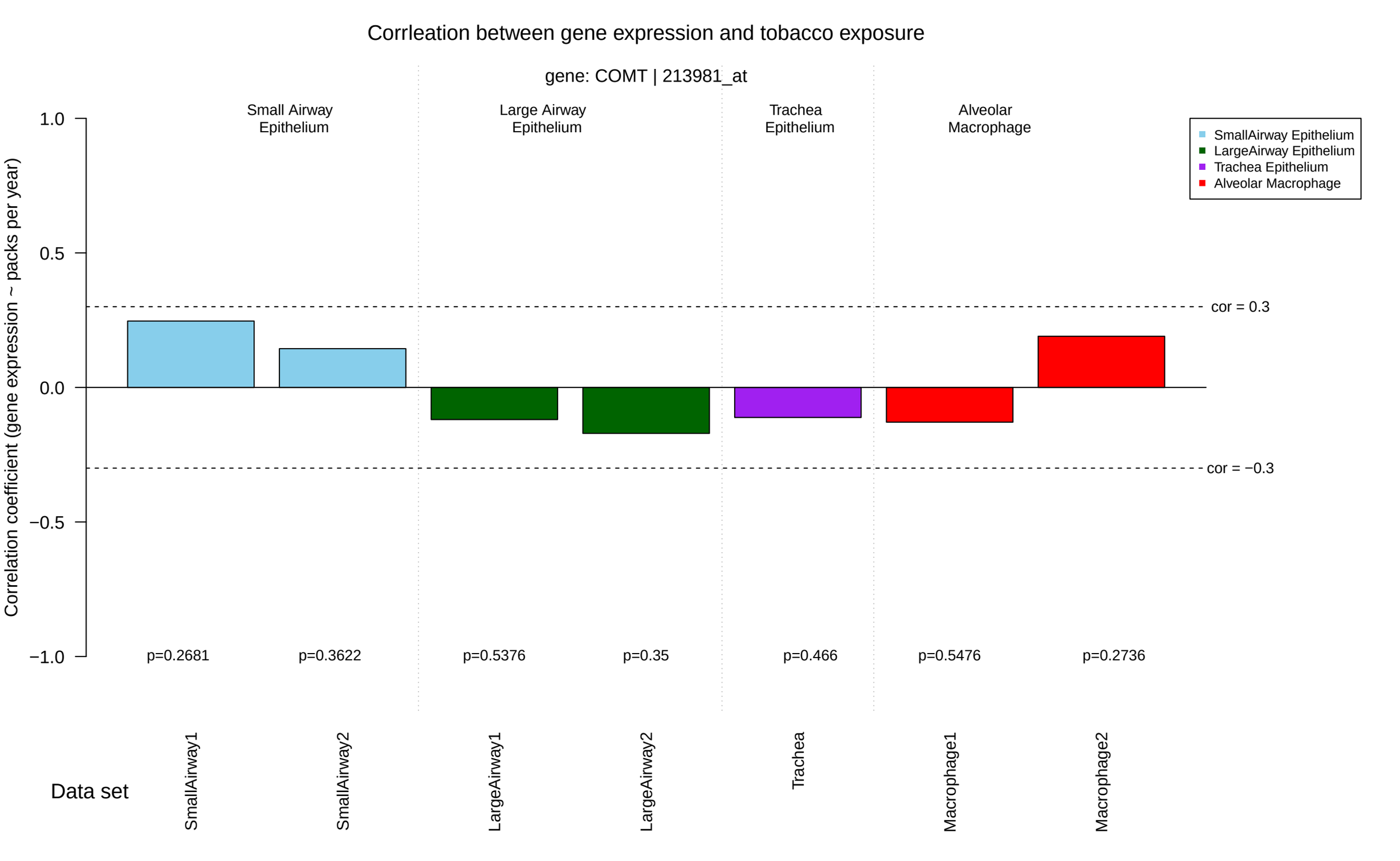

set: LargeAirway1 (p=0.9738)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7539)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8764)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7559)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2052)
COMT | 213981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COMT | 213981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2681
p=0.3622
p=0.5376
p=0.35
p=0.466
p=0.5476
p=0.2736
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2